Use work-energy to solve motion

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Choose work-energy when force and displacement are known.

SUCCESS CRITERIA - I can...

1. List all forces doing work.
2. Calculate the kinetic-energy change.
3. Check whether the result is physically possible.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A 4 N force acts forward through 6 m. Find work and show the force/displacement directions.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $W_{net} = \frac{1}{2}m(v_f^2 - v_i^2)$.
Constant mass. Use the displacement along each force component; stopping removes the initial kinetic energy.

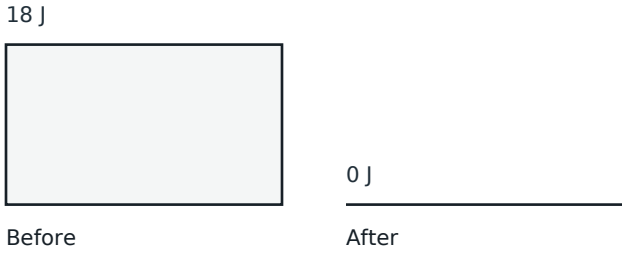
Worked reasoning: A 2 kg cart starts from rest. Constant net force 4 N acts through 3 m, doing 12 J. Then $\frac{1}{2}mv^2=12$ gives $v=\sqrt{12}\approx3.46$ m/s.

Your turn: Sketch a force diagram and an energy bar diagram for this motion. Identify where time is unnecessary.

Marzano check - circle: ① ② ③

03 Look and explain

A cart has 18 J kinetic energy and then stops. How much net work is done? Show an energy bar before and after.



Bar height represents energy. Read the value and unit above each bar.

Marzano check - circle: ① ② ③

04 Build the solution

A 4 N resistive force stops the cart from 3. Use energy removed=force×distance. Find stopping distance: 18÷4.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Use the idea

A calculation gives negative final kinetic energy after a long braking distance. What should you check?

Marzano check - circle: 1 2 3

06 Choose a method

Two carts approach: one has momentum 6 kg·m/s east and the other 4 kg·m/s west. Explain why adding their magnitudes gives the wrong total; find the signed total.

Marzano check - circle: 1 2 3

07 Exit check

A cart has 12 J initial kinetic energy and gains 8 J net work. Find final kinetic energy.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A cart needs 18 J to stop. Compare stopping distances for constant opposing forces 3 and 6 N.

Marzano check - circle: 1 2 3